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United States Patent	12408247
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Konstantopoulos; Christos et al.

Adaptive loop technique for high PSRR current regulator

Abstract

Regulator circuitry configured to manage power supply ripple using an adaptive loop gain that offers ripple rejection performance related to the ripple magnitude for a negative closed loop of the regulator circuitry. The power supply regulator circuitry of this disclosure includes an error amplifier in the closed loop and an adaptive loop gain circuitry. The adaptive loop gain circuit removes a DC component of the sensed output and feeds the sensed output, including the output ripple, to peak detector circuitry to obtain a Vpeak signal. The Vpeak signal output from the peak detector circuitry is a continuous signal that tracks the wave peak. The circuit arrangement multiplies the output of the error amplifier by the signal Vpeak resulting in improved power supply ripple rejection as the ripple amplitude increases. To avoid control signal with zero value during times that the peak-to-peak value of the sensed voltage.

Inventors: Konstantopoulos; Christos (Padua, IT), Tonazzo; Enrico (Villanova di Camposampiero, IT), Galvano; Maurizio (Padua, IT), Montoncelli; Mattia (Abano Terme, IT), Cusinato; Federico (Tombolo, IT)

Applicant: Infineon Technologies AG (Neubiberg, DE)

Family ID: 93467248

Assignee: Infineon Technologies AG (Neubiberg, DE)

Appl. No.: 18/325586

Filed: May 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240407062 A1	Dec. 05, 2024

Publication Classification

Int. Cl.: H05B45/30 (20200101); G05F1/46 (20060101); G05F1/565 (20060101); H05B45/345 (20200101); H05B45/48 (20200101)

U.S. Cl.:

CPC H05B45/345 (20200101); G05F1/467 (20130101); G05F1/565 (20130101); H05B45/48 (20200101);

Field of Classification Search

CPC: H05B (45/10); H05B (45/30); H05B (45/345); H05B (45/48); H05B (47/10); G05F (1/467); G05F (1/565)

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Primary Examiner: Vu; Jimmy T

Attorney, Agent or Firm: Shumaker & Sieffert, P.A.

Background/Summary

TECHNICAL FIELD

(1) The disclosure relates to power supply circuits and, more specifically, to techniques for managing current ripple in power supply circuits.

BACKGROUND

(2) Circuits that are driven by direct current (DC) power supplies, such as light emitting diode (LED) driver circuits and other types of circuits, may include regulator circuitry for voltage regulation and/or current regulation. The supplied DC power may be subject to undesirable power supply ripple, e.g., current ripple or voltage ripple. The regulator circuitry, in addition to maintaining the current and/or voltage within a limited range, may also help reject, attenuate, or filter the power supply ripple.

SUMMARY

(3) In general, the disclosure describes regulator circuitry configured to manage power supply ripple using an adaptive loop gain that offers ripple rejection performance related to the ripple magnitude for a negative closed loop of the regulator circuitry. Similar to other regulator circuitry, the power supply regulator circuitry of this disclosure may include an error amplifier in the closed loop to keep output power at or near the desired power level. For example, the error amplifier may output a difference between the sensed output voltage, V_{sense} , and a reference voltage, V_{ref} , and

the closed loop may operate to minimize the error, e.g., the difference between V_{sense} and V_{ref} .

(4) In addition, the regulator circuitry of this disclosure includes the adaptive loop gain circuitry which removes a DC component of the sensed output and feeds the sensed output, including the output ripple, to peak detector circuitry to obtain a V_{peak} signal. The V_{peak} signal output from the peak detector circuitry is a continuous signal that tracks the wave peak. In some examples, the peak detector circuitry may also amplify the signal. The circuit arrangement multiplies the output of the error amplifier by the signal V_{peak} . To avoid control signal with zero value during times that the peak-to-peak value of the sensed voltage, V_{sense} , may be close to zero, the circuit also adds the output of the error amplifier to the multiplied signal.

(5) In one example, this disclosure describes a circuit comprising a peak detector circuit configured to receive a sensed signal, rectify the sensed signal and emit a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; an error amplifier with a first input element, a second input element and an output element, wherein the first input terminal is configured to receive the sensed signal; and an output terminal configured to receive the output of the error amplifier added to the product of V_{peak} multiplied by the output of the error amplifier.

(6) In another example, this disclosure describes a method comprising receiving, by closed loop regulation circuitry, a sensed signal; rectifying, by an adaptive loop circuit of the closed loop regulation circuitry, the sensed signal; detecting the peak of the rectified sensed signal; emitting a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; comparing, by an error amplifier of the adaptive loop circuit, the sensed signal to a reference voltage; multiplying, by the adaptive loop circuit, an output of the error amplifier by V_{peak} , wherein the output of the error amplifier is a result of comparing the sensed signal to the reference voltage; adding, by the adaptive loop circuit, the output of the error amplifier to the product of V_{peak} multiplied by the output of the error amplifier; using, by the closed loop regulation circuitry, the sum of the output of the error amplifier and the product of V_{peak} multiplied by the output of the error amplifier to control a voltage ripple of the sensed signal.

(7) In another example, this disclosure describes a device comprising a peak detector circuit configured to receive a sensed signal, rectify the sensed signal and emit a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; an error amplifier with a first input element, a second input element and an output element, wherein the first input terminal is configured to receive the sensed signal; and an output terminal configured to receive the output of the error amplifier added to the product of V_{peak} multiplied by the output of the amplifier.

(8) The details of one or more examples of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating an example architecture for the regulator circuitry with adaptive loop gain of this disclosure.

(2) FIGS. 2A, 2B, and 2C are schematic diagrams illustrating three different examples of the plant circuit of this disclosure.

(3) FIG. 3 is a schematic diagram illustrating an example detailed architecture for the regulator circuitry of this disclosure with adaptive loop gain implemented with the plant of FIG. 2C.

(4) FIG. 4 is a graph illustrating an example of power supply rejection performance for the regulator circuitry with adaptive loop gain of this disclosure.

(5) FIG. 5 is a graph illustrating an example of performance as a function of the amplitude of the ripple the circuitry with adaptive loop gain of this disclosure.

(6) FIG. 6 is a flow chart illustrating an example operation for the circuitry of this disclosure.

DETAILED DESCRIPTION

(7) The disclosure describes regulator circuitry configured to manage power supply ripple using an adaptive loop gain that offers ripple rejection performance related to the ripple magnitude for a negative closed loop of the regulator circuitry. The regulator circuitry of this disclosure may apply electromagnetic compatibility (EMC) robustness to linear current regulators. Some applications for the circuitry of this disclosure may include driver circuits, including driver circuits for light emitting diodes, LEDs.

(8) The regulator circuitry of this disclosure may include an error amplifier in the closed loop and adaptive loop gain circuitry. The adaptive loop gain circuit may remove a direct current (DC) component of the sensed output and feed the sensed output, including the output ripple, to peak detector circuitry to obtain a V_{peak} signal. In some examples, the V_{peak} signal output from the peak detector circuitry is a continuous signal that tracks the wave peak. The circuit arrangement multiplies the output of the error amplifier by the signal V_{peak} resulting in improved power supply ripple rejection as the ripple amplitude increases.

(9) FIG. 1 is a schematic diagram illustrating an example architecture for the regulator circuitry with adaptive loop gain of this disclosure. In the example of FIG. 1, system 100 includes a circuit with adaptive loop circuitry 125, error amplifier 108, plant 102, a sense resistor, R_{sense} 122, a string of diodes 103, a supply voltage, V_s 104 with a ripple, V_{ripple} 106.

(10) Adaptive loop circuitry 125, in the example of FIG. 1, includes peak detector circuit 110, which is configured to receive sensed signal, V_{sense} 128. V_{sense} 128 may be based on an output voltage for the power supply. The circuitry of peak detector 110 may rectify the sensed signal, V_{sense} 128 and emit a continuous signal, V_{peak} 130. In some examples, V_{peak} 130 is proportional to the peak of the rectified sensed signal, and may be represented as a variable with a scalar value, K . In some examples, peak detector 110 may also amplify V_{peak} 130, e.g., in examples when the input to peak detector 110 is small amplitude, but useable signal. In some examples, peak detector 110 may selectively amplify the output, V_{peak} 130, e.g., signals above a specified threshold may not be amplified, while output signals below a specified threshold may be amplified.

(11) System 100 further includes error amplifier 108 connected to adaptive loop circuitry 125. In some examples, error amplifier 108 may be considered as a component of adaptive loop circuitry 125. Error amplifier 108 includes a non-inverting input connected to V_{ref} 120. An inverting second input element for error amplifier 108 may connect to plant 102, and receive sensed signal, V_{sense} 128.

(12) Adaptive loop circuitry 125 also includes an output terminal 132 configured to receive the output 134 of error amplifier 108. Output 134 is also added to the product of V_{peak} 130 multiplied by output 134 of error amplifier 108. Output 132 connects to terminal P3 of plant 102.

(13) In some examples, supply voltage V_s 104 connects to ground 124 and to diode string 103. The supply voltage may have a voltage ripple, modeled by V_{ripple} 106, which connects between V_s 104 and diode string 103. In the example of system 100, diode string 103 includes three diodes, D1, D2 and D3 connected in series. In other examples, diode string 103 may include any number of diodes, and in some examples may include more than one string of series connected diodes, with each string connected in parallel (not shown in FIG. 1).

(14) Diode string 103 connects to terminal P1 of plant 102. Plant 102 may receive a current, e.g., I_{sense} 126 from diode string 103. Terminal P2 of plant 102 connects to ground 124 through R_{sense} 122. Plant 102 may comprise a three-port block that acts as the channel of the regulator circuit of system 100. P1 is the input of the channel, P2 the output and P3 the control input. The negative loop of the regulator reduces the output ripple.

(15) The sense voltage, V_{sense} 128 is the voltage across R_{sense} 122. Peak detector 110 also receives V_{sense} 128. In some examples, the reference voltage V_{ref} 120 is subtracted from V_{sense} 128.

(16) In operation, the circuitry in system **100** may limit the current ripple, e.g., from Vripple **106**, delivered by the LED drivers derived from the variation of the supply line, Vs **104**. The architecture forms a negative loop such that error amplifier **108** minimizes the error, and the LED current is set according to

$$(17) I_{sense} = \frac{V_{sense}}{R_{sense}}$$

where the value of Rsense **122** may be a resistor that is defined by the designer and based on the application for the circuit.

(18) The arrangement of adaptive loop circuitry **125**, implements an adaptive loop gain so that as the power supply rejection is improved, the circuitry of system **100** may further reduce the ripple of the load current, e.g., Isense **126**. The presented architecture depicted in FIG. **1** is based on the adaptive loop gain, of the regulator, which is set linear dependent on the magnitude of the input ripple, from Vripple **106**, such that the adaptive loop gain, increases as the amplitude of the input ripple increases. In this manner the dynamic higher loop gain may improve the power supply rejection of the LED driver.

(19) Power supply rejection can be measured by the power supply rejection ratio (PSRR) under ac analysis. The inverse or reciprocal of PSRR may be represented in Equation 1:

$$(20) PSRR^{-1} = \frac{V_{sense[ac]}}{V_{ripple[ac]}} \quad [1]$$

(21) The output from terminal P2, Vsense **128**, which may include a ripple, is fed to peak detector **110**, which outputs Vpeak **130**. In some examples, peak detector **110** may include circuitry with a half wave rectifier. In some examples the DC component may be removed from Vsense **128**, e.g., by subtracting Vref **120**. As noted above, Vpeak **130** may be a continuous signal proportional to the peak of the rectified sensed signal, Vsense **128**. Vpeak **130** contains no small signal information, thus may be regarded as a variable, K. Since the DC component for Vpeak **130** is proportional to the Vsense.sub.peak which is the peak amplitude of Vsense, then the K variable is also proportional described in Equation 2.

$$(22) K \propto V_{sense_{peak}} \quad [2]$$

(23) The output **134** of error amplifier **108** is multiplied by 114 with the Vpeak **130**, which may be written as A.Math.(Vref-Vsense).Math.K. In order to avoid a control signal at terminal **132** with a zero value, e.g., in the example where Vsense.sub.peak is close to zero, the output of the error amplifier **108** is also added at **116** to the multiplied signal from **114**. The control signal Vcontrol **132** after small signal analysis is described in Equation 3.

$$(24) V_{control} = A \cdot \text{Math.} (V_{ref} - V_{sense}) + A \cdot \text{Math.} (V_{ref} - V_{sense}) \cdot \text{Math.} K \quad [3]$$

where A is the gain of error amplifier **108**. Vcontrol **132** may also be written as Equation 4.

$$(25) V_{control} = (1 + K) \cdot \text{Math.} A \cdot \text{Math.} (V_{ref} - V_{sense}) \quad [4]$$

The closed loop gain of the regulator may be represented in Equation 5.

$$(26) \frac{V_{sense}}{V_{ref}} = \frac{A \cdot \text{Math.} (1 + K) \cdot \text{Math.} G_{p2p3}}{1 + A \cdot \text{Math.} (1 + K) \cdot \text{Math.} G_{p2p3}} \quad [5]$$

(27) Where G.sub.p2p3 is defined as the small signal gain between P2 and P3 ports in open for the loop configuration. The reciprocal quantity of the PSRR,

$$(28) PSRR^{-1} = \frac{V_{sense}}{V_{ripple}},$$

for the closed loop system, is described in Equation 6 where G.sub.p2p1 is the gain of P2 and P1 ports in open loop configuration.

$$(29) PSRR^{-1} = \frac{G_{p2p1}}{1 + A \cdot \text{Math.} (1 + K) \cdot \text{Math.} G_{p2p3}} \quad [6]$$

(30) The above description and equations show that the variable K is proportional to the ripple of Vsense **128**. As the ripple in Vsense **128** increases the PSRR.sup.-1 is proportionally reduced and so the power supply rejection performance is improved.

(31) FIGS. 2A, 2B, and 2C are schematic diagrams illustrating three different examples of the plant

circuit of this disclosure. The examples of FIGS. 2A-2C illustrates the three-way configuration of plant **108** described above in relation to FIG. 1. The plant can be configured either as a single transistor device plant, e.g., FIG. 2A, as a single transistor device paired with the offload resistor Roffload **210**, e.g., as shown in FIG. 2B. The plant may also be implemented as shown in FIG. 2C as a channel that consists of the main regulation device Q1, an offload resistor, Roffload **212** and a secondary device Q2 that is driven by offload driver **220**. Offload driver **220** performs regulation of the offload current. The control port P3 of FIG. 2C is connected to the control terminal of the main regulation device, transistor Q1, and the offload regulator path.

(32) For FIGS. 2A-2C, as described above in relation to FIG. 1, P1 is the input element of the channel, P2 the output element and P3 the control input element. For the example of FIG. 2A, P1 connects to the source of N-channel transistor Q1 **201**, P2 connects to the drain of Q1 **201** and P3 connects to the control terminal, or gate, of Q1 **201**.

(33) In the example of FIG. 2B, P1 connects to P2 through offload resistor, Roffload **210**. The drain-source channel of N-channel transistor Q1 **202** connects P1 and P2 in parallel to Roffload **210**. P3 connects to the gate of Q1 **202**.

(34) For FIG. 2C, the drain-source channel of N-channel transistor Q2 **206** connects in series with Roffload **212**, and P1 connects to P2 through the series connection of the drain-source channel of Q2 **206** and Roffload **212**. The drain-source channel of N-channel transistor Q1 **204** also connects P1 and P2. P3 connects to the gate of Q1 **204**. Offload driver **208** drives the gate of Q2 **206** based on the input at P3.

(35) FIG. 3 is a schematic diagram illustrating an example detailed architecture for the regulator circuitry of this disclosure with adaptive loop gain implemented with the plant of FIG. 2C. System **300** in the example of FIG. 3 illustrates one possible implementation for the circuitry of this disclosure. Plant **302** is an example of the plant of FIG. 2C and has the same function, architecture and characteristics as described above in relation to FIG. 2C. For example, plant **302** of system **300** includes terminals P1, P2 and P3, transistors Q1 **344** and Q2 **346**, along with Roffload **332** and offload driver **338**, which are connected in the same manner as described for FIG. 2C.

(36) Adaptive loop circuitry **325**, supply voltage **304**, ripple **306**, Rsense **322** and error amplifier **308** are examples of adaptive loop circuitry **125**, supply voltage **104**, ripple **106**, Rsense **122** and error amplifier **108** described above in relation to FIG. 1 and have the same connections, functions and characteristics as described for system **100**. In the example of FIG. 3, peak detector **310** is implemented as a diode connected to an RC filter. The architecture of peak detector **310** is configured to rectify sensed signal Vsense **328** and emit a continuous signal, Vpeak **330**, which is proportional to the peak of the rectified sensed signal. In some examples, peak detector **310** may also include circuitry to amplify Vpeak **330**.

(37) In some examples, the peak detector circuit, e.g., peak detector **310**, receives a signal from **312** comprising the difference of reference voltage, Vref **320** subtracted from sensed signal, Vsense **328**, e.g., from the output terminal P2 of plant **302**. Output terminal **332** of adaptive loop circuitry **325** receives the output of error amplifier **308** added by **316** to the product of Vpeak **330** and multiplied by the output of error amplifier **308** at **314**. The plant circuit, e.g., plant **302**, includes input terminal P3 connected to output terminal **332** of adaptive loop circuitry **325**.

(38) System **300** may represent an example use case that may be analyzed for adaptive loop gain based ripple improvement architecture of this disclosure. Plant **302**, as with the plant circuits of FIGS. 2A-2C, may also be referred to as a channel in this disclosure. The current regulator is accompanied with the channel, e.g., a channel defined by plant **302**, formed by the main regulation device Q1 **344**, offload resistor Roffload **332**, and the transistor Q2 **346**. The schematic of FIG. 3 includes the power source Vs **304**, connected to ground **324**, the undesired ripple, modeled by Vripple **306**, an LED string, modeled by LEDs D1-D3, and sense resistor, Rsense **322**. For system **300**, the ripple is produced in series with the voltage source, Vs **304** and is propagated to the channel, plant **302** via terminal P1. The ripple rejection improvement of system **300** is performed

by the rectification of the output ripple by peak detector **310** that produces the V_{peak} **330** signal comprising a DC component proportional to the peak-to-peak voltage of the output ripple, V_{ripple} **306**. The multiplication of signal V_{peak} **330** with the output of error amplifier **308** at **314** increases the gain of the regulation loop in proportion to the output ripple magnitude. Thus, the power supply rejection is improved according to Equation 6 above.

(39) FIG. **4** is a graph illustrating an example of power supply rejection performance for the regulator circuitry with adaptive loop gain of this disclosure. The example of FIG. **4** illustrates the inverse power supply rejection of the adaptive loop gain implementation of this disclosure of system **300** in FIG. **3** compared to a base power supply regulation circuit without the adaptive loop circuitry of this disclosure. For example, a base power supply regulation loop may include an error amplifier that provides feedback to a plant, e.g., as in FIG. **1**, but without the peak detector output multiplied by the error amplifier output of the adaptive loop circuitry of this disclosure (not shown in FIG. **1** or **4**).

(40) The example of FIG. **4** uses a modeled one-volt power supply peak-to-peak ripple, e.g., for V_{ripple} **306** described above in relation to FIG. **3**. Curve **402** shows the performance of a 12V base circuit and curve **404** shows the improvement for a 12V circuit with the adaptive loop circuitry of this disclosure. Similarly, curve **406** shows the performance of a 24V base circuit and curve **408** shows the improvement for a 24V circuit with the adaptive loop circuitry of this disclosure.

(41) As shown in FIG. **4**, in case of the adaptive loop implementation, the ripple that is measured in V_{sense} may be attenuated by as much as 15 dB more than the base solution in the tested use case. Moreover, the adaptive loop mechanism of this disclosure may also have an advantage over other types of regulation circuitry in that the circuitry of this disclosure operates independent of the supply voltage magnitude that may force the operational regions of the devices **Q1** and **Q2** of plant **302** in FIG. **3** to change.

(42) FIG. **5** is a graph illustrating an example of performance as a function of the amplitude of the ripple the circuitry with adaptive loop gain of this disclosure. In the presented invention the ripple rejection performance is a function of the amplitude of the ripple at a certain frequency as described in Equation 6, above. FIG. **5** illustrates the power supply rejection ratio for base and adaptive loop implementation for a supply voltage, V_s , of 12 V, with a modeled input ripple of frequency at 613 kHz and a varying peak-to-peak amplitude. As shown, the power supply rejection **502** is a function of the ripple amplitude due to the particular adaptive loop gain architecture of this disclosure, while the base implementation **504** remains steady over the V_{ripple} sweep. In this manner, the adaptive loop gain architecture of this disclosure may provide advantages over other types of regulation circuits, as can be seen by the measurement of the $PSRR_{sup.-1}$ observed behavior of the $PSRR_{sup.-1}$ under varying V_{ripple} .

(43) FIG. **6** is a flow chart illustrating an example operation for the circuitry of this disclosure. The blocks of FIG. **6** will be described in terms of the example of FIG. **3**, unless other noted.

(44) As described above in relation to FIG. **3**, system **300**, which, along with system **100** of FIG. **1**, may be described as including closed loop regulation circuitry, may receive sensed signal, V_{sense} **328** from terminal **P2** of plant **302** (**90**). V_{sense} **328** may include a ripple, modeled by V_{ripple} **306**.

(45) Adaptive loop circuit **325** of the closed loop regulation circuitry, may rectify the sensed signal. V_{sense} **328** (**92**). In system **300**, peak detector **310** includes rectification circuitry, e.g., a diode, configured to rectify the received sensed signal. Peak detector **310** may also include circuitry configured to detect the peak of the rectified sensed signal. As described above in relation to FIGS. **1** and **3**, peak detector **310** may also emit a continuous signal, e.g., V_{peak} **330**, that is proportional to the peak of the rectified sensed signal.

(46) Error amplifier **308** of the adaptive loop circuit receives V_{ref} **320** at the non-inverting input and V_{sense} **328** at the inverting input in the example of FIG. **3**. Error amplifier **308** compares the sensed signal to the reference voltage and outputs A_x ($V_{ref} - V_{sense}$), where A_x is the gain of error amplifier **308** (**94**). Adaptive loop circuit **325** multiplies the output of the error amplifier by V_{peak}

330, at **314 (96)**.

(47) As described above in relation to FIGS. **1** and **3**, adaptive loop circuit **325** also adds the output of the error amplifier at **316** to the product of V_{peak} **330** multiplied by the output of the error amplifier (**97**) to avoid a zero input to the control input terminal **P3** of plant **302**. The closed loop regulation circuitry uses the sum of the output of the error amplifier and the product of V_{peak} **330** multiplied by the output of the error amplifier to control voltage ripple of the sensed signal (**98**). As described above in relation to FIGS. **4** and **5**, as the ripple increases, the gain increases, and results in better ripple control, when compared to other arrangements of regulator circuits.

(48) The disclosure may also be understood by referring to the following clauses.

(49) Clause 1: A circuit comprising a peak detector circuit configured to receive a sensed signal, rectify the sensed signal and emit a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; an error amplifier with a first input element, a second input element and an output element, wherein the first input terminal is configured to receive the sensed signal; and an output terminal configured to receive the output of the error amplifier added to the product of V_{peak} multiplied by the output of the error amplifier.

(50) Clause 2: The circuit of clause 1, wherein the second input of the error amplifier is configured to receive a reference voltage.

(51) Clause 3: The circuit of any of clauses 1 and 2, wherein the peak detector circuit is further configured to receive a signal comprising the difference of a reference voltage subtracted from the sensed signal.

(52) Clause 4: The circuit of any of clauses 1 through 3, wherein the peak detector circuit is configured to amplify V_{peak} .

(53) Clause 5: The circuit of any of clauses 1 through 4, wherein the peak detector circuit comprises a diode and an RC filter.

(54) Clause 6: The circuit of any of clauses 1 through 5 further comprising a sense resistor, wherein the sensed signal is a voltage across the sense resistor.

(55) Clause 7: The circuit of any of clauses 1 through 6, further comprising a plant circuit, wherein the plant circuit comprises an output terminal configured to output the sensed signal, and wherein the plant circuit comprises an input terminal configured to connect to the output terminal of the adaptive loop circuitry.

(56) Clause 8: The circuit of clause 7, wherein the circuit is configured to control a power supply ripple for a light emitting diode (LED) circuit.

(57) Clause 9: The circuit of clause 8, wherein the input terminal for the plant circuit is a first input terminal, the plant circuit further comprising a second input terminal, and wherein the second input terminal is configured to connect to the LED circuit.

(58) Clause 10: The circuit of any of clauses 7 through 9, wherein the plant circuit comprises an offload resistor.

(59) Clause 11: A method comprising receiving, by closed loop regulation circuitry, a sensed signal; rectifying, by an adaptive loop circuit of the closed loop regulation circuitry, the sensed signal; detecting the peak of the rectified sensed signal; emitting a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; comparing, by an error amplifier of the adaptive loop circuit, the sensed signal to a reference voltage; multiplying, by the adaptive loop circuit, an output of the error amplifier by V_{peak} , wherein the output of the error amplifier is a result of comparing the sensed signal to the reference voltage; adding, by the adaptive loop circuit, the output of the error amplifier to the product of V_{peak} multiplied by the output of the error amplifier; using, by the closed loop regulation circuitry, the sum of the output of the error amplifier and the product of V_{peak} multiplied by the output of the error amplifier to control a voltage ripple of the sensed signal.

(60) Clause 12: The method of clause 11, further comprising, subtracting the reference voltage from sensed signal before rectifying the sensed signal.

- (61) Clause 13: The method of any of clauses 11 and 12, further comprising amplifying V_{peak} before emitting V_{peak} .
- (62) Clause 14: The method of any of clauses 11 through 13, wherein the adaptive loop circuit comprises a diode and an RC filter.
- (63) Clause 15: The method of any of clauses 11 through 14, wherein the closed loop regulation circuitry comprises a sense resistor, and wherein the sensed signal is a voltage across the sense resistor.
- (64) Clause 16: The method of any of clauses 11 through 15, further comprising, controlling a power supply ripple for a light emitting diode (LED) circuit.
- (65) Clause 17: The method of clause 16, wherein the closed loop regulation circuitry further comprises a plant circuit, the method further comprising outputting, by the plant circuit the sensed signal; and receiving, by the plant circuit, the sum of the output of the error amplifier and the product of V_{peak} multiplied by the output of the error amplifier from the adaptive loop circuit.
- (66) Clause 18: The method of clause 17, further comprising, receiving, by the plant circuit, an electrical current from the LED circuit.
- (67) Clause 19: The method of any of clauses 17 and 18, wherein the plant circuit comprises an offload resistor.
- (68) Clause 20: A device comprising a peak detector circuit configured to receive a sensed signal, rectify the sensed signal and emit a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; an error amplifier with a first input element, a second input element and an output element, wherein the first input terminal is configured to receive the sensed signal; and an output terminal configured to receive the output of the error amplifier added to the product of V_{peak} multiplied by the output of the amplifier.
- (69) Various examples of the disclosure have been described. These and other examples are within the scope of the following claims.

Claims

1. A circuit comprising adaptive loop circuitry, wherein the adaptive loop circuitry comprises: a peak detector circuit configured to receive a sensed signal, rectify the sensed signal, and emit a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; an error amplifier with a first input element, a second input element, and an output element, wherein the first input element of the error amplifier is configured to receive the sensed signal; and an output terminal configured to receive an output of the error amplifier added to a product of V_{peak} multiplied by the output of the error amplifier.
2. The circuit of claim 1, wherein the second input element of the error amplifier is configured to receive a reference voltage.
3. The circuit of claim 1, wherein the peak detector circuit is further configured to receive a signal comprising the difference of a reference voltage subtracted from the sensed signal.
4. The circuit of claim 1, wherein the peak detector circuit is configured to amplify V_{peak} .
5. The circuit of claim 1, wherein the peak detector circuit comprises a diode and an RC filter.
6. The circuit of claim 1 further comprising a sense resistor, wherein the sensed signal is a voltage across the sense resistor.
7. The circuit of claim 1, further comprising a plant circuit, wherein the plant circuit comprises an output terminal configured to output the sensed signal, and wherein the plant circuit comprises an input terminal configured to connect to the output terminal of the adaptive loop circuitry.
8. The circuit of claim 7, wherein the circuit is configured to control a power supply ripple for a light emitting diode (LED) circuit.
9. The circuit of claim 8, wherein the input terminal for the plant circuit is a first input terminal, the plant circuit further comprising a second input terminal, and wherein the second input terminal is

configured to connect to the LED circuit.

10. The circuit of claim 7, wherein the plant circuit comprises an offload resistor.

11. A method comprising: receiving, by closed loop regulation circuitry, a sensed signal; rectifying, by an adaptive loop circuit of the closed loop regulation circuitry, the sensed signal; detecting a peak of the rectified sensed signal; emitting a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; comparing, by an error amplifier of the adaptive loop circuit, the sensed signal to a reference voltage; multiplying, by the adaptive loop circuit, an output of the error amplifier by V_{peak} , wherein the output of the error amplifier is a result of comparing the sensed signal to the reference voltage; adding, by the adaptive loop circuit, the output of the error amplifier to the product of V_{peak} multiplied by the output of the error amplifier; using, by the closed loop regulation circuitry, a sum of the output of the error amplifier and a product of V_{peak} multiplied by the output of the error amplifier to control a voltage ripple of the sensed signal.

12. The method of claim 11, further comprising, subtracting the reference voltage from sensed signal before rectifying the sensed signal.

13. The method of claim 11, further comprising amplifying V_{peak} before emitting V_{peak} .

14. The method of claim 11, wherein the adaptive loop circuit comprises a diode and an RC filter.

15. The method of claim 11, wherein the closed loop regulation circuitry comprises a sense resistor, and wherein the sensed signal is a voltage across the sense resistor.

16. The method of claim 11, further comprising, controlling a power supply ripple for a light emitting diode (LED) circuit.

17. The method of claim 16, wherein the closed loop regulation circuitry further comprises a plant circuit, the method further comprising: outputting, by the plant circuit the sensed signal; and receiving, by the plant circuit, the sum of the output of the error amplifier and the product of V_{peak} multiplied by the output of the error amplifier from the adaptive loop circuit.

18. The method of claim 17, further comprising, receiving, by the plant circuit, an electrical current from the LED circuit.

19. The method of claim 17, wherein the plant circuit comprises an offload resistor.

20. A device comprising adaptive loop circuitry, wherein the adaptive loop circuitry comprises: a peak detector circuit configured to receive a sensed signal, rectify the sensed signal, and emit a continuous signal, V_{peak} , wherein V_{peak} is proportional to the peak of the rectified sensed signal; an error amplifier with a first input element, a second input element, and an output element, wherein the first input element of the error amplifier is configured to receive the sensed signal; and an output terminal configured to receive an output of the error amplifier added to a product of V_{peak} multiplied by the output of the amplifier.
